ME228 · FINAL PROJECT · 2026

Data-driven fatigue strength prediction & inverse alloy design for low-alloy steels.

Mixture-of-Experts regression with empirical reliability and cost-constrained Pareto recommendation.

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PROBLEM & MOTIVATION

Fatigue dominates failure, but testing is destructive and slow.

- Fatigue accounts for **80–90 %** of mechanical structural failures.
- Rotating-bending tests run >10^7 cycles per specimen — destructive, slow, expensive.
- Handbook S–N curves cover only catalogued grades; new compositions need extrapolation.

TWO COUPLED ENGINEERING QUESTIONS

FORWARD Will this alloy + heat-treatment survive the design load?

INVERSE What is the cheapest alloy that meets the load with a stated safety margin?

GOAL *Replace handbook lookup with a calibrated forward predictor and a cost-aware inverse recommender — one pipeline, both questions.*

DATASET & KEY OBSERVATION

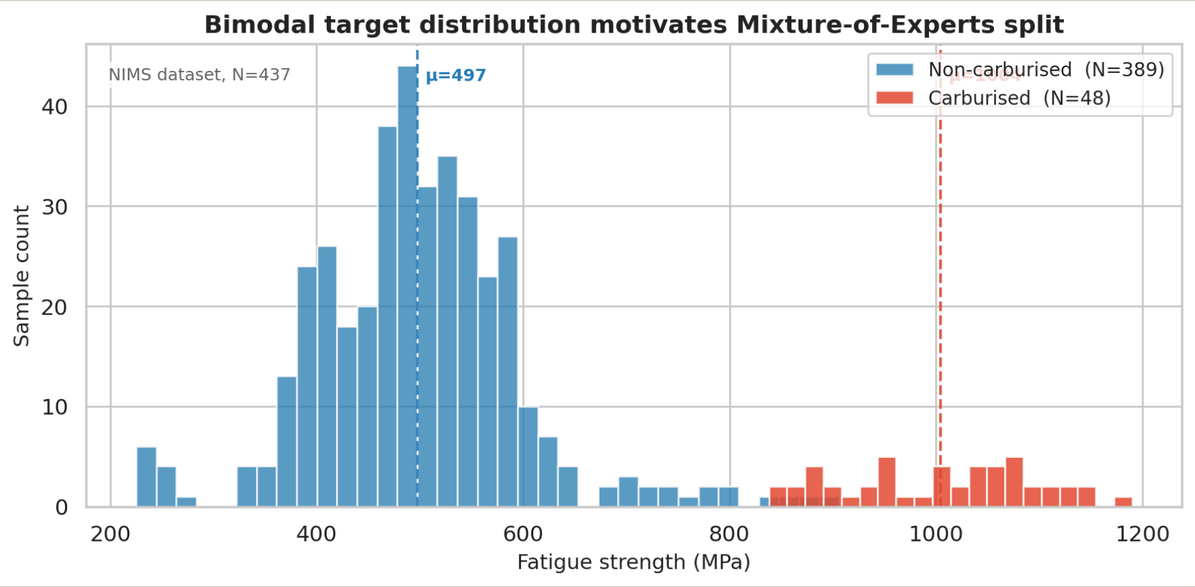
Bimodal target → split the model.

SAMPLES	INPUT FEATURES
437	26

NON-CARBURISED	CARBURISED (CT = 930 °C)
389 · 225–906 MPa	48 · 838–1190 MPa

- Composition (9): C, Si, Mn, P, S, Ni, Cr, Cu, Mo · Heat-treatment (8) · Inclusions (8) · target Fatigue (MPa).
- Two physically distinct strengthening mechanisms: case-hardening vs. through-hardening.

EMPIRICAL FATIGUE DISTRIBUTION



THROUGH-HARDENED (NC, N=389) CASE-HARDENED (C, N=48, CT=930 °C)

TRIGGER *A single global model averages two regimes and inflates regime-specific uncertainty — Mixture-of-Experts is justified by the data, not by taste.*

ARCHITECTURE

One hard gate, two regime-specific experts.

Input — composition + heat-treatment (26 features)

Regime gate *CT == 930 °C ?*

no →

Non-carburised expert
XGBoost · 12 features
19.85 ±2.44 MPa
5-fold CV-RMSE · N = 389

yes →

Carburised expert
Random Forest · 10 features
40.50 ±13.60 MPa
5-fold CV-RMSE · N = 48

↓ predicted fatigue $\hat{F}$ + uncertainty σ ↓

$$PoF_{NC} = CDF_{OOF}(\sigma_{app} - \hat{F})$$
empirical, residuals fail Shapiro

$$PoF_{C} = \Phi((\sigma_{app} - \hat{F})/\sigma)$$
Gaussian, residuals pass

Forward report $\hat{F}$ · FoS · PoF · 95 % PI

Inverse loop sample → hull-guard → Pareto (cost, PoF)

WHY MOE *The split does not win on pooled accuracy — it wins on regime-specific σ, which is what trustworthy PoF reporting actually needs.*

METHODS & HEADLINE METRICS

Pipeline discipline, then numbers.

FEATURE & HYPERPARAMETER PIPELINE

- **VIF filter** — drop the largest VIF iteratively until all < 10. Removes 6 features per regime.
- **SHAP scout** — shallow RF on VIF survivors; keep top-k by mean |SHAP|. `k = 12` NC, `k = 10` C.
- **Leak-free** — both run *inside* each fold via a custom `VIFThenShapSelector` transformer; test data never touches selection.
- **Optuna TPE** — 60 trials, 5-fold CV objective, per regime.
- **Holdout** — 20 % stratified by fatigue quintile, never touched during tuning.

FINAL SELECTED FEATURES

NC Cr · TT · C · Mn · THT · Mo · Si · RedRatio · P · S · dA · Cu

C P · RedRatio · Ct · Mo · S · Mn · Cu · Dt · TT · QmT

HEADLINE METRICS

REGIME	CV-RMSE (MPA)	HOLDOUT RMSE	R^2
Non-carburised	19.85 ± 2.44	19.19	0.970
Carburised	40.50 ± 13.60	28.47	0.860

NC R^2 (HELD-OUT)
0.97

C R^2 (HELD-OUT)
0.86

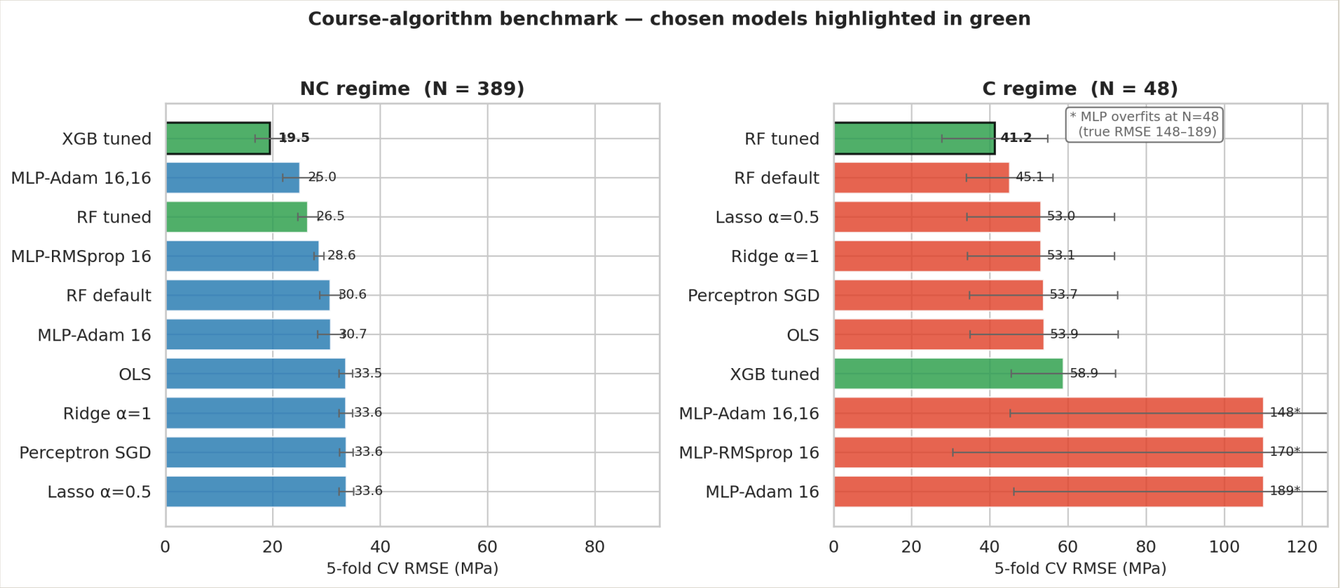
Holdout matches CV — no overfit to the fold structure.

DISCIPLINE *Leak-free VIF+SHAP inside each CV fold, Optuna tuning, evaluation on a held-out 20 % the model has never seen.*

COURSE-ALGORITHM BAKE-OFF

The chosen pair wins both regimes.

5-FOLD CV RMSE (MPA) · ME228 PALETTE



Tuned XGB / RF (green) win both regimes against tuned defaults across the four-family palette: XGBoost, Random Forest, Ridge, kNN(k=7). MLPs are also shown for the C regime to illustrate the N ≥ 10 d_{VC} failure mode.

MARGIN OVER BASELINES

TUNED DEFAULTS · 4-FAMILY PALETTE

FAMILY	NC	C
▶ **XGBoost**	23.47	59.42
▶ **Random Forest**	25.76	46.33
Ridge	36.53	75.81
kNN (k=7)	70.98	73.73

C: TUNING LIFT, XGB → RF

59.4 → 40.5 MPa

Tuned RF beats default XGB by 32 % on N = 48.

VERDICT *The chosen pair (XGBoost / RF) is empirically the best legal choice from the ME228 palette under our protocol.*

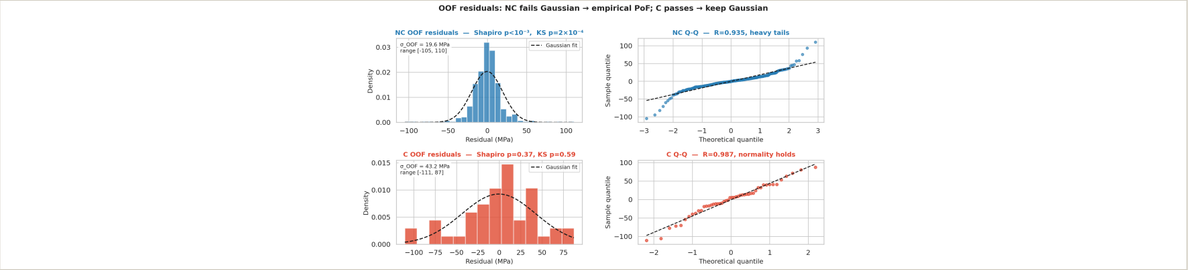
RELIABILITY — EMPIRICAL POF

Gaussian PoF is invalid for non-carburised residuals.

NORMALITY OF OOF RESIDUALS

REGIME · SOURCE	STD (MPA)	SHAPIRO P	KS P
NC · OOF	19.65	$< 10^{-3}$	2×10^{-4}
C · OOF	43.18	0.37	0.59

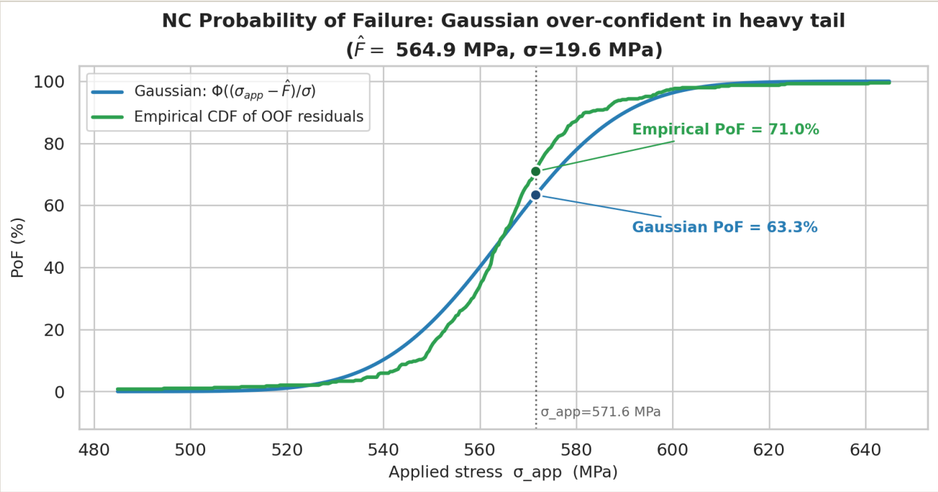
NC residuals: σ = 19.65 MPa, range [−104.9, +109.9] — heavy-tailed and asymmetric. C residuals: σ = 43.18 MPa, range [−110.5, +87.2] — Gaussian fit holds.



EMPIRICAL POF (NC)

PoF_{emp} = empirical CDF of OOF residuals at σ_{app} - $\hat{F}$ · no Gaussian assumption.

CANONICAL EXAMPLE · Σ_{APP}=590, $\hat{F}$=585.0 MPA



POF — GAUSSIAN

60.0 %

POF — EMPIRICAL

66.6 %

+6.5 pp shift · empirical CDF over OOF residuals captures the heavy tail Φ misses.

INVERSE DESIGN & COST FIX

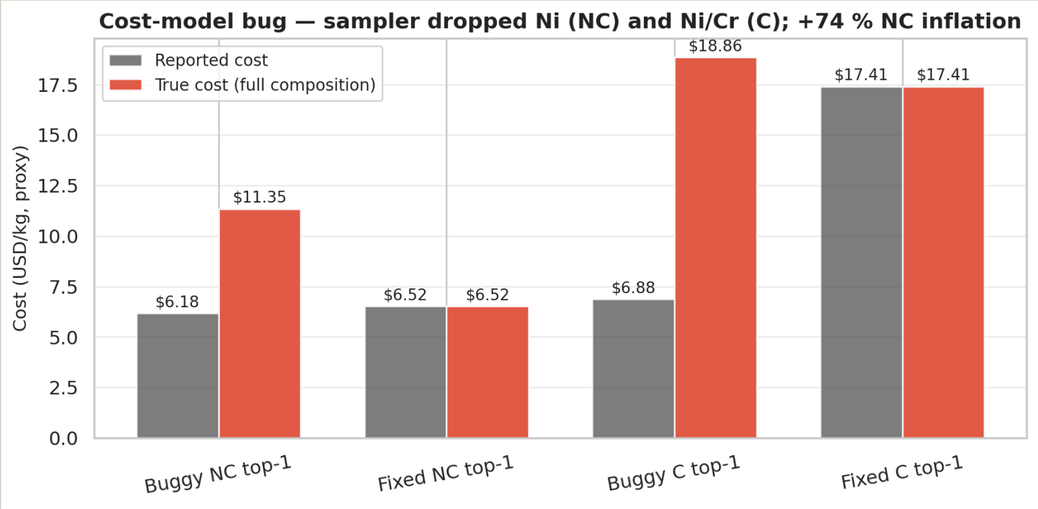
Cheapest viable alloy under FoS and PoF caps.

- **Neighbourhood sampler** — perturb real training points, δ = 0.20 σ̂ per feature, clipped to per-feature bounds.
- **kNN hull guard** — reject candidates whose mean 5-NN distance exceeds the p99 of training self-distances.
- **Cost model** — Mo 30, Ni 14, Cr 10, Cu 9, Mn 1.8, Si 1.2, C 0.5 USD/kg + carburising time.
- **Output** — Pareto front in (cost, PoF), top-K cheapest feasible alloys, bootstrap-stable across seeds.

OBSERVATION — SCHEMA DISCIPLINE IN THE CANDIDATE VECTOR

An early sampler carried only the regime feature subset — cost-only columns (e.g. Ni) defaulted to zero, biasing the cost surface low by ~74 %. We now propagate the full cost-relevant schema through the candidate vector; numbers below reflect the corrected pipeline.

COST SURFACE · PARTIAL-SCHEMA VS. CORRECTED (USD/KG)



NC TOP-1 · Σ_{APP}=500

$6.52 /kg

F̃ 755.5 MPa · Si-strengthened

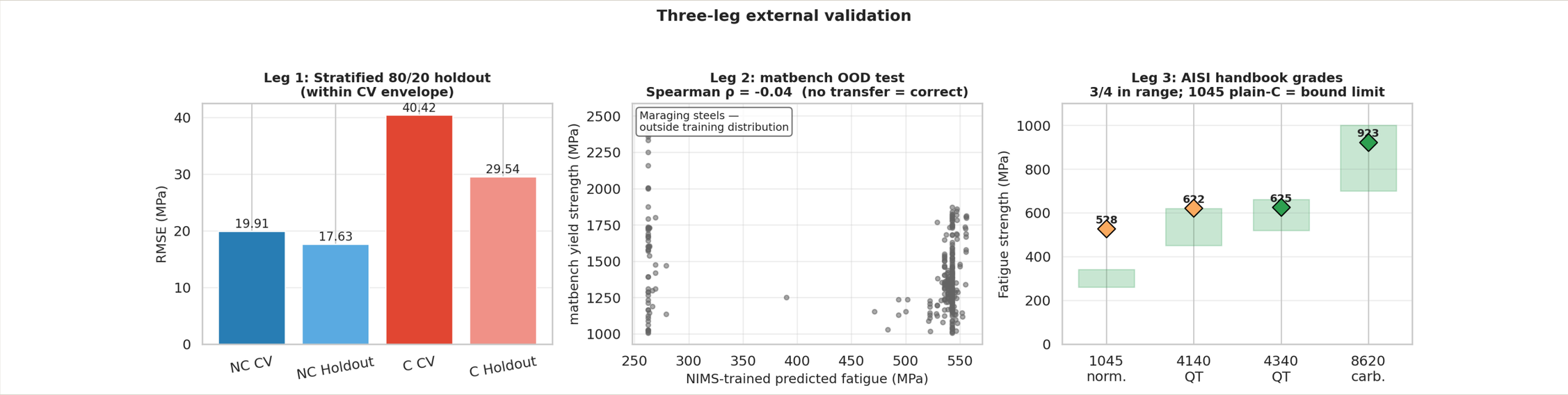
C TOP-1 · Σ_{APP}=700

$17.41 /kg

F̃ 1015.6 MPa · Cr+Mo

EXTERNAL VALIDATION

Three independent legs define the applicability boundary.



LEG 1 · STRATIFIED HOLDOUT

NC R^2 = 0.97, C R^2 = 0.86. Holdout RMSE 19.19 / 28.47 MPa — inside the CV envelope.

LEG 2 · MATBENCH_STEELS (OOD)

Spearman ρ = −0.04, p = 0.44 — a designed null. Confirms the alloyed-Fe-C-Mn-Cr-(Mo, Ni) boundary.

LEG 3 · CANONICAL AISI GRADES

4140 / 4340 / 8620 in handbook range; 1045 +188 MPa over. Deployment gate: `Cr+Ni+Mo ≥ 0.1 wt%`.

TAKEAWAY *The model is auditable inside the alloyed regime and explicitly fenced outside it — matbench and AISI 1045 mark the wall.*

LIMITATIONS · FUTURE WORK · CODE & REFERENCES

What we know we don't know — and where to push next.

LIMITATIONS

- **Plain-carbon out of scope.** AISI 1045 over by 188 MPa; matbench ρ = −0.04. Gated to `Cr+Ni+Mo ≥ 0.1 wt%`.
- **Carburised N = 48.** Uncertainty band itself uncertain — ~50–100 more samples halves it.
- **Inverse-design = band, not recipe.** Top-1 cost stable across seeds; composition Jaccard 0.31 / 0.42.
- **NC near the data hull.** p99 → p95 cuts feasible 463 → 17, lifts cost $6.52 → $9.67/kg.

FUTURE WORK

- Wang et al. 2025 broad-steel corpus (3 612 samples) — addresses carburised-N ceiling.
- Soft regime routing — nitriding, induction, dual-phase.
- Conformal prediction · supplier-quoted cost surfaces.

CODE & REPRODUCIBILITY

GITHUB

`github.com/RudraKhandelwal/ME228`

Pipeline · inverse loop · 3-leg validator · Streamlit GUI.

REFERENCES

1. Agrawal et al. *Data science techniques to predict fatigue strength of steel*. Integ. Mater. Manuf. Innov. 3(1), 2014. [NIMS]
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3. Quraishy & Kundu. *Fatigue of a Broad Range of Steels via ML*. J. Mater. Eng. Perform., 2025.
4. Budynas & Nisbett. *Shigley's Mechanical Engineering Design*, 11e, 2020. [AISI]

Thank you — questions?

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